

AWS Console, IAM & Security

Session 3 – Cloud Access
AWS Cloud Club Peshawar

Muhammad Taha

AWS Cloud Club Peshawar Workshop

Before we start, make sure you have:

- An **email address** for sign-up.
- If you have a **virtual card** (SadaPay/NayaPay), we can use Free Tier; otherwise we'll use **AWS Educate**.
- Your **phone** ready for MFA (Google Authenticator) later.

Interactive Question (Start Strong)

Question: If you wanted to build the next Netflix, where would you store the videos?

- Servers in your room?
- Google Drive?
- The cloud?

Point: Big apps use cloud infrastructure. (Netflix is a famous AWS customer.)

What is Cloud Computing? (Simple)

- **On-demand** IT resources over the Internet.
- **Pay-as-you-go**: pay only for what you use.
- Access computing resources (servers, storage, databases, software) over the internet instead of owning and maintaining physical hardware.

Why Cloud? (1 Slide)

- No servers to buy or maintain.
- Scale up/down anytime.
- Faster to build and deploy projects.

Analogy: Buying a server is like buying a whole car — expensive upfront, you maintain it, fix it, park it. Cloud is like Uber — you pay only when you need a ride. That's AWS.

What is AWS?

Amazon Web Services (AWS) is a cloud platform that provides:

- **Compute** (servers), **storage**, **databases**, and more.
- **Pay-as-you-go** pricing.
- Global infrastructure (regions around the world).

Overview of AWS (Simple)

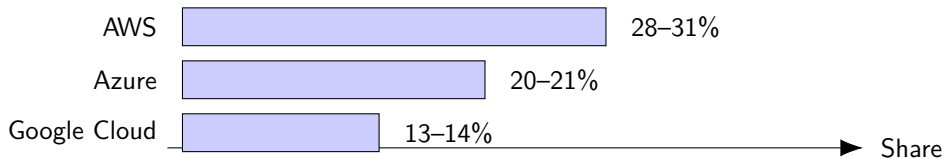
- AWS offers a **huge range of services**: compute, storage, networking, databases, and much more.
- **Pay-as-you-go**: pay only for what you use (good for cost control).
- Started in **2006**; early core services included **S3** (storage) and **EC2** (compute).
- Today AWS has **200+** services (many sources say **240+** as of early 2026).

What Makes AWS Special?

- **Scalability:** grow or shrink resources anytime.
- **Global reach:** many regions across the world.
- **Reliability:** multiple Availability Zones per region.
- **Security:** IAM, encryption, and many security controls.

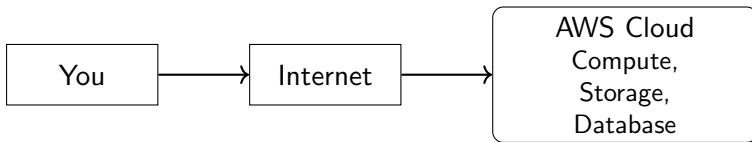
Cloud Market (Big 3) – Simple Diagram

Approx market share (Q4 2025 – Q1 2026):



Note: Numbers are approximate (commonly reported by sources like Statista / industry reports).

You and the Cloud – Diagram



You connect over the internet to AWS; AWS provides compute, storage, and many other services.

Account Creation (Main Path: Free Tier)

We will use Free Tier as the main path today.

- ➊ Go to aws.amazon.com/free → **Create a Free Account**.
- ➋ Verify email, enter contact details.
- ➌ Add card details (**PAN/Tax ID: enter 0 if asked**).
- ➍ Phone verification (SMS/Call) and choose **Basic** support.

Plan B: If your account isn't verified today, watch a friend and follow along. **Mention:** AWS Educate (no card) is another option, but today we use one main path to avoid confusion.

Free Plan vs Paid Plan (2026 Overview)

Free Plan (learning focus):

- You get **signup credits** (around \$100 + extra by doing activities).
- Lasts up to **6 months** or until credits finish.
- Access to many services, but some **features and sizes are limited**.
- Good for labs, practice, and this workshop.

Paid Plan (full access):

- You still get credits at the start, but account **does not auto-expire**.
- Access to **all** services and bigger instance types.
- Pay-as-you-go when credits are used up.

Core AWS Services (What You Must Know Today)

EC2 Virtual server (compute).

S3 File storage (buckets, objects).

IAM Users and permissions (security).

Billing/Budgets Cost control (alerts and budgets).

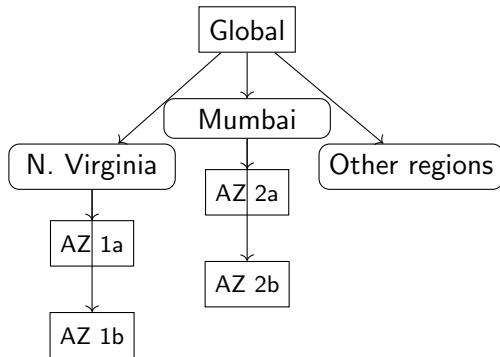
Simple analogies:

- **EC2** = *Cyber café*: rent a computer by the hour (in Amazon's data center).
- **S3** = *Warehouse*: a bucket is a shelf, objects are boxes. Store anything!
- **IAM** = *Hospital keys*: determines who gets which key to which room (doctor vs janitor).
- **Budgets** = *Bill alarm*: set it at \$0.01; if you spend 1 cent, you get an email.

Regions & Availability Zones – Diagram

Regions are physical locations (e.g. us-east-1, ap-south-1).

AZs are data centers inside a region (redundancy, lower latency).



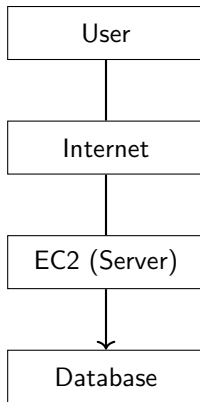
Region matters: closer region usually = lower latency (the 'Speed of Light' problem).

Analogy (The Neighborhood): Region = a City (like Islamabad/Karachi). AZ = different

The Speed of Light Problem (Edge Locations)

- **Latency:** Distance = Delay. If a user in Peshawar plays on a server in America, there will be lag.
- **Edge Locations (The Cache):** If 1,000 people in Peshawar watch the same viral video, AWS does not fetch it from America 1,000 times!
- It stores a copy nearby so the next 999 people get it instantly.
- **Analogy:** Think of how fast Netflix or TikTok loads popular videos.

Example Web App on AWS (Simple)



AWS gives you the infrastructure to build real applications.

Console Tour (What to Look For)

In the AWS Console, always find these 3 things:

- **Search bar** (type: EC2, S3, IAM, Billing).
- **Region selector** (top-right: e.g. N. Virginia us-east-1).
- **Account menu** (top-right) for security settings.

The Console Trap (Regional vs Global)

Beginner Warning: Some services are Global, some are Regional!

- **IAM is Global:** You create a user once, and they work everywhere.
- **EC2 is Regional:** If you launch a server in Mumbai, you will **not** see it if your console is set to N. Virginia.

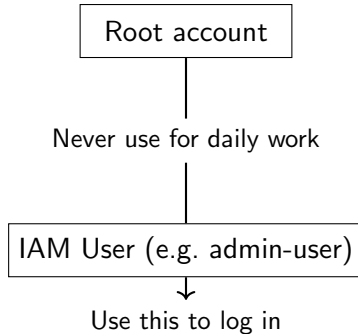
Rule: If you log in and your servers are missing, check your Region!

Part B: The “Root” Danger

Analogy: Root is like the master key to an entire building. You don't carry it around every day — you keep it in a safe and use a normal key for daily use. **What we do:**

- 1 Go to IAM Dashboard.
- 2 Create User → Name: `admin-user`.
- 3 Permissions: Attach **AdministratorAccess** policy.
- 4 Download **.csv** credentials. (*Store safely; never share or put on GitHub.*)
- 5 Log out of Root. Log in as the IAM User.

Root vs IAM User – Diagram



What is IAM?

IAM helps you **securely control access** to AWS resources.

- Manage **users, roles, and permissions**.
- Define **who can access what** in your AWS account.

Root account: Created by default. **Free service, global**. Do not use or share it for daily work.

IAM: Users, Groups, Roles, Policies (Simple)

- **User:** One person's account (like student1).
- **Group:** A team of users (like 'developers') with shared permissions.
- **Policy:** A rule — what is allowed or denied.
- **Role:** Temporary permission given to an AWS service.

Simple rule: Put users in a Group → Give the Policy to the Group. Done.

MFA (Multi-Factor Authentication)

Extra layer of security: you need two things to log in.

- **Something you KNOW** → your password.
- **Something you HAVE** → your phone (Google Authenticator 6-digit code changing every 30s).

Analogy: Your ATM card needs both the card AND the PIN. One alone is not enough. MFA is the same idea for your AWS account.

Ways to Access AWS (Simple)

Console Website (browser) – easiest for beginners.

CLI Terminal commands – good for automation later.

SDK/APIs Use AWS in your code (Python/Java/etc.).

Today we use the **Console**.

IAM Best Practices (Must Follow)

- ❶ Avoid using **root** except for setup.
- ❷ Use **MFA** on root.
- ❸ Use **groups** for permissions.
- ❹ Never share **passwords** or **access keys**.

Part C: Global Infrastructure Tour

- **Regions:** Switch between `us-east-1` (Virginia) and `ap-south-1` (Mumbai).
- **Services:** Locate EC2 (Compute) and S3 (Storage) in the console.
- **Budgeting:** Billing Dashboard → Budgets → Create a budget for **\$0.01**. (Crucial.)
- **Monitoring:** Show Billing Dashboard cost overview.

Part D: MFA Setup

- Download **Google Authenticator** on your phone.
- Link it to your **AWS Root Account** (not the IAM user).
- MFA adds a second layer of security: something you know (password) + something you have (phone).

Part E: One Simple Deploy (Optional)

Live demo (recommended): S3

- S3 → Create bucket → Upload file.
- Show the object URL.

If time: EC2

- EC2 → Launch instance → t2.micro.
- Use **User data** to auto-install Apache and a simple page.
- Then **Stop** the instance so you don't keep paying.

EC2 User Data (Apache Demo) – Optional

Example user data script (Amazon Linux): `#!/bin/bash`

```
sudo yum update -y
sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
echo "<html><h1>Welcome to Apache on Amazon Linux!</h1></html>" \
  | sudo tee /var/www/html/index.html
```

Paste this in the **User data** box when launching EC2. Then open `http://<EC2-public-ip>/` in your browser to see the page.

AWS for Beginners – The Golden Rules

- ❶ **Never commit keys** – Never upload `AWS_ACCESS_KEY` to GitHub. Bots will find it and mine crypto on your bill.
- ❷ **Turn it off** – If you aren't using an EC2 instance, Stop it.
- ❸ **Budget alarm** – Set a billing alarm for \$0.01. If you spend 1 cent, you get an email.
- ❹ **Region matters** – Check your region (e.g. `us-east-1` for tutorials, or `me-south-1` / `ap-south-1` for speed).

GitHub Assignment (Part 1: Practice Labs)

During or after the session, complete these labs:

- 1 Create your own AWS account (Free Plan) and **log in**.
- 2 Go to **IAM** and find where users are listed.
- 3 Create an **S3 bucket** and upload one file.
- 4 Set a **\$0.01 budget** and check where your current cost is shown.
- 5 (Bonus) Launch a tiny **EC2** instance and then **Stop** it.

These small steps will make AWS feel familiar very quickly.

GitHub Assignment (Part 2: Submission)

Task: Prove you have set up your secure AWS environment!

- 1 Take a **screenshot** of the AWS Console logged in as an **IAM user** (not Root) in the **N. Virginia** region.
- 2 Create a **GitHub Repository**.
- 3 Upload the screenshot to the repository.
- 4 Write a **good README.md** explaining your practice labs (AWS basics, IAM, Regions).
- 5 **Submit the GitHub Repo link** so we can track and mark your progress!

Practice Q&A (Common Questions)

- **Q: Difference between Root and IAM user?**

A: Root has unlimited power (use only for setup). IAM user is for daily use. If Root gets hacked, you lose everything.

- **Q: What happens if I forget to stop my EC2 instance?**

A: It keeps charging you! That is why we set a \$0.01 budget alarm.

- **Q: Why use IAM Groups?**

A: Easily manage permissions for multiple users at once (e.g., 10 developers).

- **Q: Is AWS free?**

A: There is a Free Tier with limits (750 EC2 hrs/mo, 5GB S3). Exceeding limits means charges apply.

How to Start Learning AWS (Simple Roadmap)

- 1 Learn **EC2** (servers).
- 2 Learn **S3** (storage).
- 3 Learn **IAM** (security).
- 4 Build a small project (example: **host a static website on S3**).

Quick Cheat Sheet

- **Cloud** = Renting computers over the internet (Uber analogy).
- **AWS** = World's biggest cloud platform, 200+ services.
- **EC2** = Virtual computer/server (Cyber café analogy).
- **S3** = File storage like Google Drive for apps (Warehouse analogy).
- **IAM** = Who can access what (Hospital key analogy).
- **Root** = God Mode — lock it away, never use daily.
- **MFA** = Password + phone code (ATM analogy).
- **Region** = City — **AZ** = Building inside city.
- **Always set \$0.01 budget alarm.**
- **NEVER put AWS keys on GitHub** — bots will find them!